

LEZIONE

6

DUE EQUAZIONI

$$p(x) = 0$$

$$\begin{aligned} p(x) &\geq 0 \quad \text{or} \quad p(x) \leq 0 \end{aligned}$$

$$\begin{aligned} p(x) &> 0 \\ p(x) &< 0 \end{aligned}$$

IN UNA EQUAZIONE
DIVERGENTE

SE SPOSTO UNA QUANTITÀ
DA DX A SX
DX ⁰ A DX

CAMBIA SEGNO

SE DIVIDO UN'EQUAZIONE
O DISEQUAZIONE

PER UNA QUANTITÀ

$\begin{cases} \text{positiva} & \text{NON CAMBIA SEGNO} \\ \text{negativa} & \text{CAMBIA SEGNO} \end{cases}$

NON DIVIDERE PER X

$$-3x + 4 > 0$$

$$-3x + 4 > 0 \rightarrow 3x < 4 \rightarrow x < 4/3$$

$$2x + 1 > \pi \Rightarrow x > \frac{\pi - 1}{2}$$

per x

$$x < 0$$

$$2x - 1 > 2x$$

$$-1 > 0 \rightarrow x > \frac{1}{2-2}$$

$$x > 2$$

$$2x + 2x > 2$$

$$2 < 2 \rightarrow \frac{1}{2-2}$$

$$x(2-2) > 1$$

PARABOLA

$$ax^2 + bx + c = 0$$

$$a \neq 0$$

$$x_{1/2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a > 0$$

$$\uparrow \downarrow$$

$$\Delta < 0$$

$$\uparrow \downarrow$$

$$\Delta = 0$$

$$\uparrow \downarrow$$

$$\Delta > 0$$

$$\uparrow \downarrow$$

$$\Delta < 0$$

$$\uparrow \downarrow$$

$$\Delta = 0$$

$$\uparrow \downarrow$$

$$\Delta > 0$$

es
 $x^2 + x + 2 < 0$ ha soluzione?

~~1/4/7/0~~ $4 - 1 < 0 \rightarrow x < \frac{1}{4}$

es 2

$x^2 + 2 \leq 3x$

$x^2 + 2 \leq 3x$

$x^2 - 3x + 2 \leq 0 \quad 2 \leq x \leq 2$

es 3

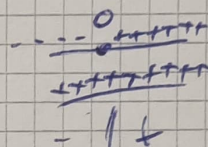
$x^3 + 2x^2 + x = 0$

$x(x^2 + 2x + 1) = 0 \quad x(x+1)^2 = 0$

es 4

$x^3 + 2x^2 + x > 0$

$x(x+1)^2 > 0 \rightarrow$



es 5

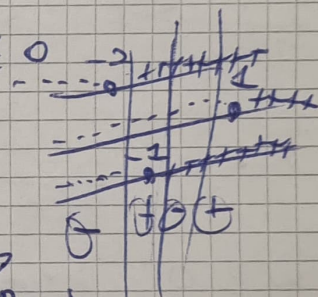
$x^3 + 2x^2 - x - 2 \leq 0$

RACCOMANDO TOIACB

$x^2(x+2) - (x+2) \leq 0$

$(x+2)(x^2-1) \leq 0$

$(x+2)(x-1)(x+1) \leq 0$



2
cambio

R.

$x \leq -2$

$-1 \leq x \leq 1$

$x+2 \geq 0$

$x-1 \geq 0$

es 6

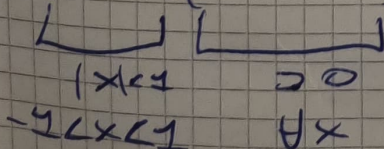
$x^4 + x^3 - 3 - 4 < 0$

$x^3(x+1) - (x+1) < 0$

$(x+1)(x^3-1) < 0$

$(x+1)(x-1)(x^2+x+1) < 0$

$(x^2-1)(x^2+x+2) > 0$



$x > \sqrt{2}$
 $x \leq \sqrt{2}$

es 7

$x^4 - x^2 - 2 \geq 0 \quad [y = x^2]$

$y^2 - y - 2 \geq 0$

$\frac{1 \pm \sqrt{1+8}}{2}$

$\frac{1 \pm 3}{2}$

$x^2 \leq -1$

$x \geq 2$

EX 2 RAZONABLE FROM RAZONABLE

$$\frac{p(x)}{q(x)} = 0 \Leftrightarrow \begin{cases} p(x) = 0 \\ q(x) \neq 0 \end{cases}$$

ex 1

$$\frac{x-2}{x^2-x-2} = 0 \Rightarrow x^2-x-2 \neq 0 \begin{pmatrix} x \neq 2 \\ x \neq -2 \end{pmatrix}$$

ex 2

$$\frac{1}{x-1} > \frac{1}{x+2} \quad \frac{1}{x-1} - \frac{1}{x+2} > 0 \quad \frac{x+2-(x-1)}{(x-1)(x+2)} > 0 \quad \frac{3}{(x-1)(x+2)} > 0$$

$$x^2-1 > 0 \quad x > 1 \vee x < -1$$

ex 3

$$\frac{|x|}{2-x} > 1 \quad x \neq 2 \quad \frac{|x|}{1-x} - 1 > 0 \quad \frac{|x| - 1 + x}{1-x} > 0$$

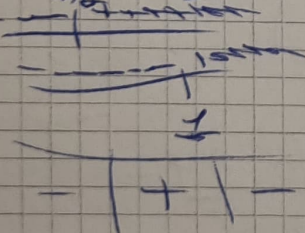
non pondra a variable

$$\cancel{x < 0} \quad \cancel{\frac{-1}{1-x} > 0} \rightarrow \cancel{1-x < 0} \rightarrow \cancel{x > 1}$$

$$x \geq 0 \quad \frac{2x-1}{1-x} > 0$$

$$2x-1 > 0 \Rightarrow x > \frac{1}{2}$$

$$1-x > 0 \Rightarrow x < 1$$



$$\frac{1}{2} < x < 1$$